Appendix A

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A.1 SUPPLEMENTARY TABLES FOR CHAPTER 2

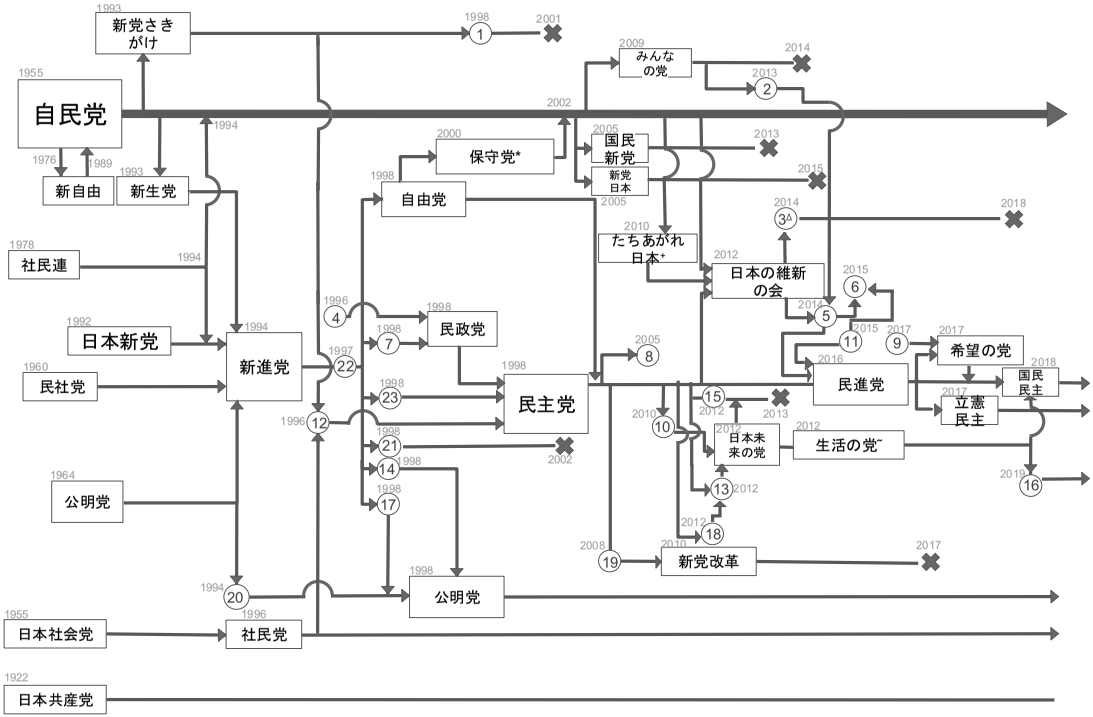

Figure A.1 Tree depicting the transformation of Japan's party system, 1955–2018 (Japanese names)
See Figure A.2 for a key to the English and Japanese names of parties represented by numbers in the circles.

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1. Sakigake/さきがけ
2. Unity Party/ 結いの党
3. Party for Future Generations/ 次世代の党
4. Sun Party/ 太陽党
5. Japan Innovation Party/ 維新の党
6. Initiatives for Osaka/ 大阪の維新の党
7. Voice of the People/ 国民の声
8. New Party Daichi/ 新党大地
9. Tomin First no Kai/都民ファースト
10. Genzei Nippon/ 減税日本
11. Vision of Reform/ 改革結集の会
12. Democratic Party of Japan/民主党
19. Japan Renaissance Party/ 改革クラブ
20. Komei/公明
21. Japan Renaissance Party/ 改革クラブ
22. Kaito/ 解党
23. New Fraternity Party/ 新党友愛
13. People's Life First/ 国民の生活が第一
14. New Party Peace/ 新党平和
15. Green Wind/ みどりの風
16. Reiwa Shinsengumi/ れいわ新選組
17. Dawn Party/ 黎明クラブ
18. Kizuna Part/ 新党きづな

*The Conservative Party changed its name to New Conservative Party (保守党 → 保守新党)
太陽の党

+ The Sunrise Party of Japan changed its name to the Sunrise Party (たちあがれ日本→太陽の党)
Δ The Party for Future Generations changed its name to the Party for Japanese Kokoro (次世代の党→日本のこころを大切にする党)
~ The People's Life Party changed its name to People's Life Party & Taro Yamamoto and Friends
(生活の党→生活の党と山本太対と仲間達)

Figure A.2 Key to the English and Japanese names of parties represented by numbers in Figures 2.1 and A.1.

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A.2 NOTES ON DATA COLLECTION, CHAPTERS 6–8

To examine the hypotheses offered in Chapter 4, I put together a comprehensive new dataset on the universe of Japanese municipalities that existed in any form between 1980 and 2014. This data is an expanded, cleaned-up version of data collected for Catalinac, Bueno de Mesquita and Smith (2020). Putting it together entailed extracting data from three data sources, cleaning the data, compiling it into a single dataset, and constructing variables to test my hypothesis. This was a difficult undertaking for several reasons, one of which was the thousands of changes to the borders of municipalities and electoral districts that took place during this period, which were all dealt with differently across the data sources. I strove to create a dataset that accounted for every single one of these changes with unique municipality and district codes, which would facilitate the use of state-of-the-art regression specifications to test my hypotheses. Below, I explain how the dataset was made.

The first step was to compile voting data. For data on voting behavior in the twelve Lower House elections held between 1980 and 2014, I relied on the municipality-level election results compiled by Mizusaki (2014). Known as the JED-M Sosenkyo data, this is not a compiled dataset, but a series of text files, each of which pertains to a single electoral district in a single election. Electoral districts were multimember prior to 1994, electing between three and five candidates, and single member after 1994. Each file contains district-specific information, including the number of seats available, the number of municipalities in the district, the voting population, the number of votes cast, the number of valid votes cast, the number of candidates running, and the names of those candidates and their party affiliations. Each file also contains a list of the municipalities in the district and records, for each municipality, the voting population, number of votes cast, number of valid votes cast, and number of valid votes cast for each candidate. A feature of this dataset that proved difficult is that municipalities (and electoral districts, for that matter) are identified by the *name* they had at the time of the election, not by any government-issued code. Files for elections held after 1994 attach a suffix onto the names of municipalities spanning more than one district, but files for elections held prior to 1994 do not. In these elections, I identified this small number of municipalities by extracting municipalities whose names appeared in more than one file and were located in contiguous electoral districts.[1]

[1] Note that the fourteen municipalities located on the Amami Islands, which constituted an electoral district until the 1993 election, when they were merged into Kagoshima 1, were excluded from the data.

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After extracting the municipality-level voting data, I used the Reed and Smith (2015) dataset to attach pertinent information about the identity of the candidates who ran in each electoral district in each election. Of particular interest was whether the candidate was an incumbent, the number of terms the candidate had served in the Lower House (if any), and whether the candidate had run as an independent, only to join the LDP after the election. Under Japan's old electoral system, a good number of the independents running in any given election were those who had failed to win the LDP's official endorsement prior to the election but had every intention of joining the party after the election, should they emerge victorious. Sometimes these candidates had even been fielded by an LDP faction intent on enlarging its numbers. Faction leaders would seek out districts in which the party's incumbents were all affiliated with other factions. Should there be another candidate aspiring to run there, the faction would throw its resources behind this candidate, with the idea that if she was victorious, she could join the party (and faction) after the election (Reed 2009). Japanese politics researchers regularly find that these LDP-aligned independents behave like regular LDP candidates (e.g., Ariga 2015). I also relied on the Reed and Smith (2015) data for its unique electoral district identifiers, which denote changes to district boundaries. Most of these changes occurred with the electoral reform in 1994 and with the redistricting that occurred between 2000 and 2003 and 2012 and 2014, respectively.

For NTD allocations, as well as other fiscal and demographic variables, I turned to the Nikkei Economic Electronic Databank System (or "Nikkei NEEDs" for short). Nikkei NEEDs provided data on the following municipality-level variables for the 1980–2012 period: total NTD allocation, total taxable income, population, fiscal strength, number of residents employed in primary industries, number of residents aged fifteen and below, number of residents aged sixty-five and above, and area size. I used these variables to construct the dependent variable (the per capita NTD allocation received by a municipality in the year after Lower House elections), as well as lags of the dependent variable and indicators of need in each municipality, which are standard controls in work on the political determinants of transfers in Japan. For each of the eight variables, NEEDs supplied a spreadsheet of data compiled from various government reports.[2] For NTD allocations and fiscal strength, NEEDs uses municipalities' "general account settlements," which are released after the fiscal year is over (April 1 until March 31) and municipalities have settled their

[2] A description of the data is available at https://needs.nikkei.co.jp/. I used NEEDs data from 2015.

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accounts (April to May).[3] For total taxable income, NEEDs relies on annual reports published by the Ministry of Internal Affairs and Communications.[4] For population, it relies on annual reports published by the Japan Geographic Data Center.[5] For residents in primary industries and aged fifteen and below and sixty-five and above, it uses data from censuses carried out every five years.[6] Finally, for area size (measured in kilometers squared), it relies on data from Japan's Geospatial Information Authority.[7]

The NEEDs spreadsheets contain data pertaining to all municipalities that existed in the post-2000 period. For this set of municipalities, data was typically available from 1980 until either 2012 or the fiscal year prior to the municipality ceasing to exist (due to a municipal merger). This data structure posed significant challenges. To explain, consider Municipality A, which ceased to exist in 2005 due to a merger with a neighboring municipality. The new (merged) entity is called Municipality B. In the NEEDs data, both municipalities are included and are distinguished by name and official municipality code.[8] For Municipality A, data is populated from 1980 until fiscal year 2004. For Municipality B, however, some variables are populated from 2005 onward (which corresponds to the years it exists), while other variables are populated from 1980 until 2012. Because Municipality B did not exist until fiscal year 2005, the

[3] For towns and villages, it uses the government's "Shichoson Betsu Kessan Jyokyo Shirabe." For cities and special wards, it uses data collated by the Nikkei Shimbun. Reports from the 2002 fiscal year onward are available at www.soumu.go.jp/iken/kessan_jokyo_2.html. The "fiscal strength" of a municipality is an index that reflects the proportion of the cost of services that a municipality is able to finance with its own tax revenue.

[4] The "Shicho Son Zei Kazei Jyoukyou Nado no Shirabe" report.

[5] The "Jyuumin Kihon Daichou Jinkou Youran" report.

[6] For the off-years, we took the value in the census year closest to the off-year.

[7] The "Zenkoku Todoufuken Shikuchouson Betsu Menseki Shirabe" report. This data was only available from 1998. I assigned municipalities with identical names and government codes in previous years to the area size they had in 1998.

[8] A merit of the NEEDs data is that information about earlier and later border changes can be incorporated into municipality names and codes. In my final dataset, for municipalities that experience a boundary change in 2005, for example, the municipality's name in the years prior to 2005 includes a suffix that describes, in parentheses, what happens in 2005. When it comes to the code, NEEDs assigns new codes every time a new entity is created but uses coding conventions to indicate whether a municipality is an earlier version of another municipality. For example, when a larger municipality absorbs other, smaller municipalities, the larger municipality might have the same name before and after the merger (with the suffix attached in earlier years, explaining what happens in later years) and the same code, distinguished by a decimal suffix. For example, let us say Municipality A exists until 2005, after which it absorbs a few municipalities on its border. Its name is Municipality A before and after 2005 (with the suffix attached to the name in earlier years), and its code is 100.1 before 2005 and 100 thereafter.

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variables populated from 1980 until 2004 are imputed. Using a comprehensive list of the 2,198 municipal mergers that have occurred since 2000, scripts were written to keep data for the years in which municipalities actually existed and delete data for the years in which they did not. This left me with fiscal and demographic information for almost all the Japanese municipalities that existed in any form between 1980 and 2012.[9] In the dataset, variables are populated only for the years in which a municipality existed.

After creating the dataset of fiscal and demographic variables, the next task was to merge this with the voting data. This posed another significant challenge: The voting data is compiled soon after each Lower House election and identifies municipalities by the names they had *at the time*. The NEEDs data, in contrast, was published in 2015 and identifies municipalities by name and official municipality code, but both the name and code reflect the ones used by the municipality *in the post-2000 period*. The name a municipality used in 2001, for example, can be slightly different to the name it had in 1991, not because of any changes to its border, but because it changed its name, has a name for which slightly different renderings in Japanese are possible, or experienced a population change that resulted in a change in its designation (as a city, town, or village).[10] Similarly, the official code a municipality used in 2001 can be different from the code it used in 1991, even though its borders are identical. These problems meant that I had municipality-years in the voting data that did not match municipality-years in the carefully pruned fiscal and demographic data.

To remedy this, I used a comprehensive dataset compiled by Kuniaki Nemoto, in which the municipality-years in the JED-M voting data had been matched by hand to the official codes these municipalities had at the time. Using his data, I investigated all unmatched observations (municipalities in the voting data that had not been matched to municipalities in the fiscal/demographic data). I found that the vast majority of observations had not been matched due to tiny differences in the rendering of municipality names. I also found that 158 unmatched observations pertained to municipalities that had existed in the 2003 and 2005 Lower House elections (and thus, had voting data) but had ceased to exist later

[9] We say "almost all" because NEEDs does not report data for the handful of municipalities that disappeared prior to 2000. Because these municipalities did not exist in the post-2000 period, NEEDs does not make data available for them.

[10] In Japanese, designations as cities (shi), towns (chou), and villages (son) appear as suffixes attached to municipality name. Moreover, municipalities sometimes change their names; for example, to differentiate themselves from another municipality in Japan with the same name.

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that same fiscal year due to a municipal merger (fiscal/demographic variables are not reported for municipalities in fiscal years in which they cease to exist). Through this, I was able to match almost all municipality-years for which I had voting data to municipality-years for which fiscal and demographic data existed.

The final step in the data collection was adding fiscal and demographic variables for the 2013, 2014, and 2015 fiscal years, which were not included in the NEEDs data. Fortunately, data for these years was available in the Japanese government's online statistics portal. I downloaded data on my variables of interest and merged them into the master dataset. Compared to earlier years, adding data for more recent years is straightforward due to the smaller number of municipalities, smaller number of name changes, and fewer changes to the borders of municipalities. It only required taking into account a handful of changes to municipality names and codes. The final master data contains variables pertaining to 105,353 municipality-years.

A.3 SUPPLEMENTARY MATERIAL FOR CHAPTER 6

Table A.1 *Descriptive statistics for the variables used in Chapter 6's analysis of transfer allocations to municipalities within districts, 1980–2014.*

Statistic	*N*	Mean	St. Dev.	Min	Max
Postelection Transfers (log)	29,088	−3.483	0.781	−6.526	1.448
Log(Transfers$_{m,t}$)	29,913	−3.469	0.770	−7.922	1.954
Best LDP VS$_{m,t}$	30,270	0.262	0.168	0.000	0.918
Rank (Best LDP VS$_{m,t}$)	25,653	0.500	0.306	0.000	1.000
Sum LDP VS$_{m,t}$	30,270	0.338	0.200	0.000	0.918
Rank (Sum LDP VS$_{m,t}$)	25,653	0.500	0.306	0.000	1.000
High LDP VS$_{m,t}$	30,270	0.230	0.171	0.000	0.918
Best LDP+ VS$_{m,t}$	30,270	0.272	0.165	0.000	0.918
Best Senior LDP VS$_{m,t}$	30,270	0.127	0.167	0.000	0.918
Best Non-LDP VS$_{m,t}$	30,270	0.131	0.133	0.000	0.818
Best Losing LDP VS$_{m,t}$	30,270	0.059	0.110	0.000	0.940
District Winner VS$_{m,t}$	30,270	0.281	0.144	0.001	0.918
Fiscal Strength$_{m,t}$	29,748	0.422	0.279	0.000	3.030
Dependent Population$_{m,t}$	29,021	0.364	0.047	0.025	0.642
Farming Population$_{m,t}$	29,020	0.091	0.072	0.000	0.625
Log(Population$_{m,t}$)	30,241	9.570	1.256	5.215	13.638
Log(Per Capita Income$_{m,t}$)	29,913	−0.145	0.403	−2.148	2.008
Population Density$_{m,t}$	30,241	0.734	1.796	0.001	21.237

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Table A.2 *The same specifications in Table 6.2 are presented here, with a control for the municipality-level vote share captured by the LDP's highest vote-getter in the district. The results are not explained by this rival theory.*

	Dependent variable: Postelection Transfers (log)			
	(Model 1)	(Model 2)	(Model 3)	(Model 4)
Best LDP $VS_{m,t}$	0.150**			
	(0.055)			
Rank (Best LDP $VS_{m,t}$)		0.036*		
		(0.017)		
Sum LDP $VS_{m,t}$			0.152**	
			(0.050)	
Rank (Sum LDP $VS_{m,t}$)				0.051**
				(0.018)
Log($Transfers_{m,t}$)	0.435***	0.436***	0.435***	0.436***
	(0.012)	(0.012)	(0.012)	(0.012)
High LDP $VS_{m,t}$	0.020	0.036	0.022	0.025
	(0.040)	(0.039)	(0.039)	(0.039)
Fiscal $Strength_{m,t}$	−0.011	−0.020	−0.012	−0.020
	(0.075)	(0.076)	(0.075)	(0.076)
Dependent $Population_{m,t}$	0.686*	0.588	0.694*	0.589
	(0.334)	(0.331)	(0.334)	(0.331)
Farming $Population_{m,t}$	−0.567	−0.617	−0.556	−0.614
	(0.328)	(0.329)	(0.326)	(0.328)
Log($Population_{m,t}$)	−0.515***	−0.520***	−0.512***	−0.517***
	(0.116)	(0.116)	(0.116)	(0.116)
Log(Per Capita $Income_{m,t}$)	0.028	0.025	0.030	0.028
	(0.071)	(0.072)	(0.071)	(0.071)
Population $Density_{m,t}$	0.026	0.032	0.025	0.032
	(0.075)	(0.077)	(0.075)	(0.077)
Observations	15,526	15,416	15,526	15,416
District-Year FE	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y
R^2	0.205	0.205	0.205	0.206

Robust standard errors clustered on municipality in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

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Table A.3 *The same specifications in Table 6.4, with alternative specifications of electoral support.*

	Dependent variable: Postelection Transfers			
	(Model 1)	(Model 2)	(Model 3)	(Model 4)
Sum LDP $VS_{m,t}$	0.153**			
	(0.048)			
Sum Senior LDP $VS_{m,t}$	0.028			
	(0.048)			
Sum LDP+ $VS_{m,t}$		0.180***		
		(0.049)		
Sum Non-LDP $VS_{m,t}$			0.048	
			(0.060)	
Sum Losing LDP $VS_{m,t}$				−0.251***
				(0.061)
Log($Transfers_{m,t}$)	0.435***	0.435***	0.436***	0.436***
	(0.012)	(0.012)	(0.012)	(0.012)
Fiscal $Strength_{m,t}$	−0.012	−0.012	−0.014	−0.012
	(0.075)	(0.075)	(0.075)	(0.075)
Dependent $Population_{m,t}$	0.695*	0.689*	0.691*	0.699*
	(0.334)	(0.334)	(0.333)	(0.333)
Farming $Population_{m,t}$	−0.558	−0.556	−0.589	−0.567
	(0.326)	(0.326)	(0.327)	(0.326)
Log($Population_{m,t}$)	−0.512***	−0.510***	−0.526***	−0.529***
	(0.116)	(0.116)	(0.116)	(0.116)
Log(Per Capita $Income_{m,t}$)	0.030	0.032	0.020	0.026
	(0.071)	(0.071)	(0.071)	(0.071)
Population $Density_{m,t}$	0.025	0.025	0.028	0.028
	(0.074)	(0.074)	(0.075)	(0.074)
Observations	15,526	15,526	15,526	15,526
District-Year FE	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y
R^2	0.205	0.205	0.205	0.206

Robust standard errors clustered on municipality in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

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Table A.4 *A municipality's per capita transfers in the year following Lower House elections as a cubic function of its position in a ranking constructed on the basis of electoral support, 1980–2014.*

	Dependent variable: Postelection Transfers	
	(Model 1)	(Model 2)
Rank (Best LDP $VS_{m,t}$)	−2.060*	
	(1.009)	
Rank (Best LDP $VS_{m,t}$)^2	−0.264	
	(3.235)	
Rank (Best LDP $VS_{m,t}$)^3	3.150	
	(2.586)	
Rank (Sum LDP $VS_{m,t}$)		−2.100*
		(0.970)
Rank (Sum LDP $VS_{m,t}$)^2		0.016
		(2.823)
Rank (Sum LDP $VS_{m,t}$)^3		2.837
		(2.211)
Fiscal $Strength_{m,t}$	2.395***	2.402***
	(0.612)	(0.613)
Dependent $Population_{m,t}$	21.733***	22.170***
	(4.102)	(4.109)
Farming $Population_{m,t}$	0.220	0.121
	(2.326)	(2.337)
Log($Population_{m,t}$)	−1.295***	−1.277***
	(0.264)	(0.261)
Log(Per Capita $Income_{m,t}$)	2.014	2.002
	(1.134)	(1.134)
Population $Density_{m,t}$	0.325***	0.321***
	(0.076)	(0.076)
Observations	23,909	23,909
District-Year FE	Y	Y
R^2	0.047	0.046
Joint Hypothesis Test	0.000	0.000

Robust standard errors clustered on municipality in parentheses.
$^{*}p < 0.05$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

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Table A.5 *A municipality's per capita transfers in the year following Lower House elections as a cubic function of its position in a ranking constructed on the basis of electoral support, 1980–1993.*

	Dependent variable: Postelection Transfers	
	(Model 1)	(Model 2)
Rank (Best LDP $VS_{m,t}$)	−0.653	
	(1.561)	
Rank (Best LDP $VS_{m,t}$)^2	−3.098	
	(5.323)	
Rank (Best LDP $VS_{m,t}$)^3	4.730	
	(4.345)	
Rank (Sum LDP $VS_{m,t}$)		−0.780
		(1.474)
Rank (Sum LDP $VS_{m,t}$)^2		−2.587
		(4.552)
Rank (Sum LDP $VS_{m,t}$)^3		4.157
		(3.643)
Fiscal $Strength_{m,t}$	1.272*	1.262*
	(0.522)	(0.524)
Dependent $Population_{m,t}$	23.603**	23.993**
	(8.275)	(8.384)
Farming $Population_{m,t}$	−2.207	−2.277
	(2.769)	(2.820)
Log($Population_{m,t}$)	−1.331***	−1.308***
	(0.385)	(0.376)
Log(Per Capita $Income_{m,t}$)	1.716	1.674
	(1.199)	(1.198)
Population $Density_{m,t}$	0.415***	0.403**
	(0.125)	(0.124)
Observations	15,416	15,416
District-Year FE	Y	Y
R^2	0.039	0.038
Joint Hypothesis Test	0.000	0.000

Robust standard errors clustered on municipality in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

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Table A.6 *A municipality's per capita transfers in the year following Lower House elections as a cubic function of its position in a ranking constructed on the basis of electoral support, 1996–2014.*

	Dependent variable: Postelection Transfers
Rank (Sum LDP $VS_{m,t}$)	−2.788
	(1.553)
Rank (Sum LDP $VS_{m,t}$)^2	1.723
	(4.063)
Rank (Sum LDP $VS_{m,t}$)^3	2.196
	(2.917)
Fiscal $\text{Strength}_{m,t}$	5.010***
	(1.002)
Dependent $\text{Population}_{m,t}$	30.372***
	(6.029)
Farming $\text{Population}_{m,t}$	4.003
	(3.510)
Log($\text{Population}_{m,t}$)	−1.092***
	(0.166)
Log(Per Capita $\text{Income}_{m,t}$)	3.722
	(2.149)
Population $\text{Density}_{m,t}$	0.248*
	(0.104)
Observations	8,493
District-Year FE	Y
Joint Hypothesis Test	0.000
R^2	0.084

Robust standard errors clustered on municipality in parentheses.
***$p < 0.001$.
Regression models estimated with R's plm() package.

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Table A.7 *Without controlling for district-level differences thought to influence transfers, the relationship between electoral support and transfers becomes negative and statistically significant (Models 1 and 2), 1980–2014. Similarly, there is a negative relationship between support and transfers when votes for losing LDP candidates are included in the numerator (Model 3).*

	Dependent variable: Postelection Transfers		
	(Model 1)	(Model 2)	(Model 3)
Best LDP $VS_{m,t}$	−0.039*		
	(0.018)		
Sum LDP $VS_{m,t}$		−0.054**	
		(0.017)	
All LDP $VS_{m,t}$			−0.072***
			(0.016)
Log($Transfers_{m,t}$)	0.788***	0.788***	0.788***
	(0.007)	(0.007)	(0.007)
Fiscal $Strength_{m,t}$	−0.018	−0.018	−0.016
	(0.024)	(0.024)	(0.024)
Dependent $Population_{m,t}$	0.369***	0.370***	0.360***
	(0.106)	(0.105)	(0.105)
Farming $Population_{m,t}$	−0.021	−0.014	−0.008
	(0.071)	(0.071)	(0.071)
Log($Population_{m,t}$)	−0.004	−0.004	−0.004
	(0.004)	(0.004)	(0.004)
Log(Per Capita $Income_{m,t}$)	−0.166***	−0.166***	−0.171***
	(0.020)	(0.020)	(0.020)
Population $Density_{m,t}$	0.008***	0.008***	0.007***
	(0.002)	(0.002)	(0.002)
Observations	27,225	27,225	27,225
Year FE	Y	Y	Y
R^2	0.681	0.681	0.681

Robust standard errors clustered on municipality in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

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A.4 SUPPLEMENTARY MATERIAL FOR CHAPTER 7

Table A.8 *Descriptive statistics for the variables used in Chapter 7's analysis of transfer allocations to electoral districts. Observations are tournament-possible electoral districts with LDP winners in Lower House elections, 1980–2014.*

Statistic	*N*	Mean	St. Dev.	Min	Max
Postelection Transfers (log)	1,414	−3.488	0.597	−7.071	−0.407
Fiscal Strength$_{d,t}$	1,414	0.598	0.250	0.000	1.414
Farming Population$_{d,t}$	1,414	0.043	0.034	0.0002	0.195
Dependent Population$_{d,t}$	1,414	0.334	0.060	0.007	0.483
Population Density$_{d,t}$	1,414	1.415	2.952	0.020	18.278
Log(Population$_{d,t}$)	1,414	13.168	0.498	10.950	14.630
Log(Per Capita Income$_{d,t}$)	1,414	0.038	0.413	−2.204	1.585
HI$_{d,t}$	1,414	0.145	0.153	0.002	0.978
Total Seats$_{d,t}$	1,414	2.203	1.558	1	6
People Per Seat$_{d,t}$	1,414	3.249	1.172	0.570	5.897
Log(Number of Municipalities$_{d,t}$)	1,414	2.615	0.867	0.693	4.159
LDP VS$_{d,t}$	1,414	0.342	0.102	0.090	0.677
Winning LDP VS$_{d,t}$	1,414	0.325	0.100	0.074	0.657
LDP Seats$_{d,t}$	1,414	0.819	0.252	0.200	1.000
Senior LDP Politician$_{d,t}$	1,414	0.474	0.499	0	1
Number LDP Candidates$_{d,t}$	1,414	1.629	0.922	1	5

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Table A.9 *The same models in Table 7.1, but with alternative indicators for strength of LDP support in the district. Models 1 and 3 use LDP Seats$_{d,t}$, while Models 2 and 4 use LDP VS$_{d,t}$.*

	Dependent variable: Postelection Transfers (log)			
	All electoral districts		Excludes prefectural capitals	
	(Model 1)	(Model 2)	(Model 3)	(Model 4)
HI$_{d,t}$	0.720***	0.716***	0.620***	0.619***
	(0.129)	(0.128)	(0.147)	(0.148)
Fiscal Strength$_{d,t}$	−0.253*	−0.263*	−0.264*	−0.278*
	(0.106)	(0.106)	(0.117)	(0.118)
Dependent Population$_{d,t}$	4.368***	4.405***	4.008***	4.033***
	(0.537)	(0.537)	(0.551)	(0.552)
Farming Population$_{d,t}$	3.547***	3.487***	3.391***	3.268***
	(0.867)	(0.881)	(0.918)	(0.921)
Log(Population$_{d,t}$)	−0.008	0.022	−0.080	−0.040
	(0.109)	(0.109)	(0.125)	(0.121)
Log(Per Capita Income$_{d,t}$)	0.444***	0.447***	0.400*	0.403*
	(0.123)	(0.124)	(0.158)	(0.159)
Population Density$_{d,t}$	0.031*	0.031*	0.028*	0.028*
	(0.012)	(0.012)	(0.013)	(0.013)
Log(Number of Municipalities$_{d,t}$)	0.174***	0.175***	0.152***	0.151**
	(0.043)	(0.044)	(0.046)	(0.046)
People Per Seat$_{d,t}$	−0.067	−0.077	−0.031	−0.040
	(0.040)	(0.040)	(0.042)	(0.042)
LDP Seats$_{d,t}$	−0.220		−0.183	
	(0.127)		(0.143)	
LDP VS$_{d,t}$		−0.188		−0.056
		(0.196)		(0.209)
Senior LDP Politician$_{d,t}$	−0.046	−0.047	−0.037	−0.041
	(0.029)	(0.029)	(0.030)	(0.029)
Observations	1,414	1,414	1,151	1,151
Year FE	Y	Y	Y	Y
R^2	0.407	0.405	0.392	0.391

Robust standard errors clustered on electoral district in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

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Table A.10 *The models in Table 7.1 with no controls for the strength of support for the LDP.*

	Dependent variable: Postelection Transfers (log)	
	All electoral districts	Excludes prefectural capitals
	(Model 1)	(Model 2)
$\text{HI}_{d,t}$	0.715***	0.614***
	(0.129)	(0.146)
Fiscal $\text{Strength}_{d,t}$	−0.260*	−0.272*
	(0.104)	(0.117)
Dependent $\text{Population}_{d,t}$	4.407***	4.036***
	(0.536)	(0.550)
Farming $\text{Population}_{d,t}$	3.267***	3.177***
	(0.880)	(0.932)
Log($\text{Population}_{d,t}$)	0.037	−0.033
	(0.107)	(0.120)
Log(Per Capita $\text{Income}_{d,t}$)	0.436***	0.397*
	(0.124)	(0.160)
Population $\text{Density}_{d,t}$	0.033**	0.029*
	(0.012)	(0.013)
Log(Number of $\text{Municipalities}_{d,t}$)	0.171***	0.150**
	(0.043)	(0.046)
People Per $\text{Seat}_{d,t}$	−0.078	−0.043
	(0.040)	(0.042)
Observations	1,414	1,151
Year FE	Y	Y
R^2	0.402	0.389

Robust standard errors clustered on electoral district in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

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Table A.11 *The models in Table 7.2 with an alternative control for strength of LDP support in the district (LDP Seats$_{d,t}$ instead of Winning LDP VS$_{d,t}$).*

	Dependent variable: Postelection Transfers (log)			
	All electoral districts		Excludes prefectural capitals	
	(Model 1)	(Model 2)	(Model 3)	(Model 4)
HI$_{d,t}$	1.280***	2.340***	1.206**	3.240***
	(0.272)	(0.636)	(0.423)	(0.948)
Fiscal Strength$_{d,t}$	−0.238	−0.352**	−0.174	−0.212
	(0.207)	(0.136)	(0.199)	(0.188)
Dependent Population$_{d,t}$	6.820***	0.145	3.980*	−1.028
	(1.617)	(1.046)	(1.734)	(1.303)
Farming Population$_{d,t}$	3.602**	−0.998	2.359*	−2.586
	(1.305)	(1.185)	(1.009)	(1.448)
Log(Population$_{d,t}$)	0.068	−0.644***	−0.066	−0.985***
	(0.156)	(0.152)	(0.162)	(0.258)
Log(Per Capita Income$_{d,t}$)	0.343	0.460***	−0.112	0.124
	(0.218)	(0.113)	(0.234)	(0.147)
Population Density$_{d,t}$	0.045	−0.053*	0.041	−0.037
	(0.031)	(0.021)	(0.028)	(0.046)
Log(Number of Municipalities$_{d,t}$)	0.212	0.975**	0.173	0.494
	(0.111)	(0.304)	(0.128)	(0.408)
People Per Seat$_{d,t}$	−0.092	−0.020	−0.006	−0.013
	(0.061)	(0.028)	(0.053)	(0.032)
LDP Seats$_{d,t}$	−0.293*	0.003	−0.070	0.021
	(0.136)	(0.026)	(0.128)	(0.029)
Senior LDP Politician$_{d,t}$	−0.111**	−0.004	−0.118*	0.001
	(0.038)	(0.011)	(0.046)	(0.013)
Observations	578	578	382	382
Year FE	Y	Y	Y	Y
District FE	N	Y	N	Y
R^2	0.474	0.278	0.439	0.267

Robust standard errors clustered on electoral district in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

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Table A.12 *Table 7.2 with an alternative control for strength of LDP support in the district (LDP $VS_{d,t}$ instead of Winning LDP $VS_{d,t}$).*

	Dependent variable: Postelection Transfers (log)			
	All electoral districts		Excludes prefectural capitals	
	(Model 1)	(Model 2)	(Model 3)	(Model 4)
$HI_{d,t}$	1.292***	2.326***	1.229**	3.218***
	(0.277)	(0.635)	(0.426)	(0.944)
Fiscal $Strength_{d,t}$	−0.233	−0.354**	−0.181	−0.225
	(0.204)	(0.136)	(0.200)	(0.192)
Dependent $Population_{d,t}$	6.970***	0.124	3.928*	−1.132
	(1.619)	(1.042)	(1.731)	(1.318)
Farming $Population_{d,t}$	3.585**	−1.049	2.268*	−2.762
	(1.311)	(1.181)	(1.000)	(1.478)
Log($Population_{d,t}$)	0.101	−0.646***	−0.048	−0.995***
	(0.158)	(0.152)	(0.161)	(0.259)
Log(Per Capita $Income_{d,t}$)	0.335	0.460***	−0.123	0.125
	(0.219)	(0.113)	(0.235)	(0.148)
Population $Density_{d,t}$	0.046	−0.053*	0.042	−0.040
	(0.030)	(0.021)	(0.028)	(0.047)
Log(Number of $Municipalities_{d,t}$)	0.211	0.966**	0.172	0.480
	(0.112)	(0.304)	(0.128)	(0.405)
People Per $Seat_{d,t}$	−0.102	−0.020	−0.0004	−0.010
	(0.066)	(0.028)	(0.057)	(0.032)
LDP $VS_{d,t}$	−0.306	0.035	0.065	0.092
	(0.253)	(0.076)	(0.235)	(0.085)
Senior LDP $Politician_{d,t}$	−0.119**	−0.004	−0.125**	−0.00000
	(0.039)	(0.011)	(0.046)	(0.013)
Observations	578	578	382	382
Year FE	Y	Y	Y	Y
District FE	N	Y	N	Y
R^2	0.467	0.278	0.438	0.269

Robust standard errors clustered on electoral district in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

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Table A.13 *Table 7.5 on the sample of electoral districts from the preelectoral reform period, 1980–1993*

	Dependent variables:		
	Winning LDP $VS_{d,t}$	LDP $VS_{d,t}$	LDP $Seats_{d,t}$
	(Model 1)	(Model 2)	(Model 3)
$HI_{d,t}$	−0.087	−0.044	−0.137
	(0.062)	(0.042)	(0.096)
Fiscal $Strength_{d,t}$	0.020	0.017	−0.022
	(0.026)	(0.022)	(0.043)
Dependent $Population_{d,t}$	−0.110	0.186	−0.410
	(0.248)	(0.194)	(0.392)
Farming $Population_{d,t}$	0.160	0.139	0.090
	(0.221)	(0.172)	(0.289)
Log($Population_{d,t}$)	−0.083	−0.130	−0.031
	(0.116)	(0.075)	(0.171)
Log(Per Capita $Income_{d,t}$)	−0.003	−0.011	0.013
	(0.016)	(0.014)	(0.027)
Population $Density_{d,t}$	−0.005	−0.004	−0.009
	(0.004)	(0.003)	(0.005)
Total $Seats_{d,t}$	−0.021	−0.028	−0.102*
	(0.029)	(0.020)	(0.041)
Log(Number of $Municipalities_{d,t}$)	0.0002	−0.008	−0.011
	(0.018)	(0.014)	(0.024)
People Per $Seat_{d,t}$	−0.021	0.010	−0.035
	(0.044)	(0.028)	(0.068)
Senior LDP $Politician_{d,t}$	0.038***	0.020***	0.060***
	(0.009)	(0.006)	(0.015)
Number LDP $Candidates_{d,t}$	0.063***	0.102***	0.143***
	(0.007)	(0.006)	(0.012)
Observations	586	586	586
Year FE	Y	Y	Y
R^2	0.585	0.827	0.583

Robust standard errors clustered on electoral district in parentheses.
$^{*}p < 0.05$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

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Table A.14 *Table 7.5 on the sample of electoral districts from the postelectoral reform period, 1996–2014.*

	Dependent variables:		
	Winning LDP $VS_{d,t}$	LDP $VS_{d,t}$	LDP $Seats_{d,t}$
	(Model 1)	(Model 2)	(Model 3)
$HI_{d,t}$	−0.074*	−0.047*	−0.196*
	(0.033)	(0.022)	(0.094)
Fiscal $Strength_{d,t}$	−0.043	−0.049**	−0.122
	(0.027)	(0.019)	(0.077)
Dependent $Population_{d,t}$	−0.031	0.061	−0.277
	(0.093)	(0.061)	(0.270)
Farming $Population_{d,t}$	0.606	0.289	1.042
	(0.334)	(0.212)	(0.969)
Log($Population_{d,t}$)	0.170*	0.164**	0.243
	(0.078)	(0.054)	(0.237)
Log(Per Capita $Income_{d,t}$)	0.052**	0.057***	0.123*
	(0.016)	(0.017)	(0.054)
Population $Density_{d,t}$	−0.009***	−0.009***	−0.027**
	(0.002)	(0.002)	(0.008)
Log(Number of $Municipalities_{d,t}$)	−0.002	0.001	−0.029
	(0.014)	(0.009)	(0.039)
People Per $Seat_{d,t}$	−0.062**	−0.058***	−0.081
	(0.024)	(0.016)	(0.073)
Senior LDP $Politician_{d,t}$	0.148***	0.067***	0.393***
	(0.010)	(0.006)	(0.025)
Observations	1,291	1,291	1,291
Year FE	Y	Y	Y
R^2	0.268	0.292	0.188

Robust standard errors clustered on electoral district in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

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A.5 SUPPLEMENTARY MATERIAL FOR CHAPTER 8

Table A.15 *Descriptive statistics pertaining to the variables used in Chapter 8's analyses of turnout in Japanese municipalities in Lower House elections, 1980–2014.*

Statistic	*N*	Mean	St. Dev.	Min	Max
$\text{Turnout}_{m,t}$	30,270	0.729	0.105	0.364	0.981
$\text{Turnout}_{m,t-1}$	26,588	0.742	0.099	0.399	0.981
Single LDP Winner $\text{VS}_{m,t}$	30,270	0.362	0.233	0.000	0.970
Rank(Single LDP Winner $\text{VS}_{m,t}$)	25,653	0.500	0.306	0.000	1.000
All LDP Winners $\text{VS}_{m,t}$	30,270	0.458	0.255	0.000	0.974
All LDP Candidates $\text{VS}_{m,t}$	30,270	0.540	0.186	0.000	0.974
Single LDP Loser $\text{VS}_{m,t}$	30,270	0.081	0.149	0.000	0.968
Single Non-LDP+ Winner $\text{VS}_{m,t}$	30,270	0.161	0.174	0.000	0.958
Single DPJ Winner $\text{VS}_{m,t}$	30,270	0.039	0.136	0.000	0.906
$\text{Margin}_{d,t}$	30,270	0.111	0.143	0.0001	0.906
Fiscal $\text{Strength}_{m,t}$	29,748	0.422	0.279	0.000	3.030
Dependent $\text{Population}_{m,t}$	29,021	0.364	0.047	0.025	0.642
Farming $\text{Population}_{m,t}$	29,020	0.091	0.072	0.000	0.625
Log($\text{Population}_{m,t}$)	30,241	9.570	1.256	5.215	13.638
Log(Per Capita $\text{Income}_{m,t}$)	29,913	−0.145	0.403	−2.148	2.008
Population $\text{Density}_{m,t}$	30,241	0.734	1.796	0.001	21.237

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Table A.16 *Turnout rates as a cubic function of the degree to which a municipality concentrated votes on a single LDP winner relative to other municipalities in its district-year, for all Japanese municipalities, 1980–2014.*

	Dependent variable: Turnout$_{m,t}$
Rank(Single LDP Winner VS$_{m,t}$)	−0.044***
	(0.009)
Rank(Single LDP Winner VS$_{m,t}$)^2	0.064**
	(0.022)
Rank(Single LDP Winner VS$_{m,t}$)^3	−0.0004
	(0.015)
Fiscal Strength$_{m,t}$	−0.032***
	(0.004)
Dependent Population$_{m,t}$	0.432***
	(0.022)
Farming Population$_{m,t}$	0.066***
	(0.012)
Log(Population$_{m,t}$)	−0.022***
	(0.001)
Log(Per Capita Income$_{m,t}$)	0.034***
	(0.004)
Population Density$_{m,t}$	−0.003***
	(0.001)
Observations	24,320
District-Year FE	Y
Joint Hypothesis Test	0.000
R^2	0.520

Robust standard errors clustered on municipality in parentheses.
$^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

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